Virtual Machine Deployment Guide

Oracle VM VirtualBox - Technical Operation Documentation

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Purpose: To provide standardized, repeatable procedures for deploying and managing virtual machines using Oracle VM VirtualBox across multiple operating system environments.

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Content

Introduction

Virtualization

Oracle VM VirtualBox

Host System Requirements

- Hardware Requirements
- Software Requirements

Network Configuration (Role-Based Virtual Networking)

- Network Adapter Types
- Role-Based Design Principles
  - Roles
- Configuration Steps
- Operational Considerations

Naming Conventions (Reference Guidance)

- Virtual Machine Name (VirtualBox Manager)
- Hostname (Guest OS)

Post-Installation ISO and Boot Order

Snapshot Management

VM: Windows Server 2019 Standard (Desktop Experience)

- Operating System Details
- System Requirements
- VM Creation
  - Resource Allocation & Firmware Configuration
- Operating System Installation
- Initial Login
- Shutting Down

VM: Windows 11 Client (Unattended Installation)

- Operating System Details
- System Requirements
- VM Creation Process
  - Resource Allocation & Firmware Configuration
- Operating System Installation
- Shutting Down

VM: Kali Linux 2026.2 Desktop (attended Installation)

- Operating System Details
- System Requirements
- ISO Integrity Verification
- VM Creation Process
  - Resource Allocation & firmware configuration
  - Additional Settings
- Operating System Installation
- Initial Login
  - Post-Installation Configuration
- Shutting Down

Introduction

This document provides a comprehensive operational guide for deploying virtual machines using Oracle VM VirtualBox. It offers clear, repeatable procedures for creating and managing virtual machine instances across multiple operating systems, including server, client, and Linux platforms. The objective is to establish a consistent deployment methodology that can be reused and referenced as new virtual machine environments are introduced.

Virtual machine deployments are foundational to modern IT operations, enabling testing, development, training, and production simulation without dedicated physical hardware. This guide focuses on practical execution, covering configuration decisions, resource allocation, installation methods, and post-deployment verification. Each procedure is explicit and unambiguous, enabling administrators of varying experience to follow successfully.

The guide maintains a consistent structure across all deployments for clarity, scalability, and ease of maintenance. As new virtual machines or operating systems are introduced, you can append additional sections while following the same standardized format.

Virtualization

Virtualization creates a virtual representation of computing resources, including operating systems, storage, networking, and hardware components. Multiple isolated operating systems (virtual machines) can run concurrently on a single host, each with its own allocated resources and environment.

At the core of virtualization is the hypervisor, which manages and allocates physical resources to virtual machines. The hypervisor lets multiple operating systems coexist without interfering with one another.

Importance of Virtualization

- **Efficient resource utilization:** Multiple workloads can share the same physical resources, reducing hardware costs.
- **Isolation and fault containment**: Each VM is separated, improving stability and security.
- **Flexibility and scalability:** You can create, modify, duplicate, or remove VMs without changing physical infrastructure.
- **Rapid recovery and portability**: You can export, import, or migrate VMs between hosts.

Virtualization simulates complex environments with limited hardware and underpins modern cloud computing. Mastering virtual machine deployment builds transferable skills for on-premises, hybrid, and cloud environments.

Oracle VM VirtualBox

Oracle VM VirtualBox is cross-platform, open-source virtualization software supporting Windows, macOS, Linux, and Solaris hosts. Guest operating systems can include Windows, Linux distributions, and others.

Key Characteristics

- **Broad OS support**: You can deploy multiple versions of Windows Server, Windows client OS, and Linux distributions in the same environment.
- **Feature-rich**: Snapshots, virtual networking modes, shared folders, and extensible storage allow rollback, recovery, and network simulation.

Purpose in This Guide

Oracle VM VirtualBox is the standardized platform for all documented VM deployments. Its consistent interface, support for manual and unattended installations, and compatibility with industry-standard VM formats ensure repeatable deployment and reconstruction.

Host System Requirements

Before deploying virtual machines, the host system must meet specific requirements to ensure stable and reliable operation. The host system is the physical machine on which Oracle VM VirtualBox is installed, and its capabilities directly affect the performance and scalability of all virtual machines running on it.

Hardware Requirements

- **CPU**:
    - 64-bit processor
    - Hardware virtualization support (Intel VT-x or AMD-V) enabled in BIOS/UEFI
    - Recommended minimum: 4 physical CPU cores
    - Hyper-Threading (HT) / SMT supported.
- **Memory**:

    - Minimum: 8 GB RAM
    - Recommended: 16 GB RAM
    - Allocate memory carefully to avoid overcommitting resources.
- **Storage**:
    - Minimum: 200 GB of available disk space
    - Use an SSD for storage to improve VM performance.

Software Requirements

- Install Oracle VM VirtualBox and the matching Extension Pack.
- ISO installation media for each OS.
- Fully updated host OS with administrative privileges.
- Disable conflicting virtualization platforms.

Network Configuration (Role-Based Virtual Networking)

Network configuration depends on the virtual machine's functional role, not on operating system type (server vs client).

Network Adapter Types

- **NAT (Network Address Translation)**: Allows a virtual machine to access external networks (including the internet) through the host system's network connection. The virtual machine is isolated from the physical network and is not directly reachable by other devices on the local network.
- **Internal Network**: Creates a private, isolated network that exists only between virtual machines explicitly connected to the same internal network name. It provides no external connectivity unless you configure routing through another virtual machine.
- **Bridged Adapter**: Connects the virtual machine directly to the host's physical network interface. The virtual machine appears as a separate device on the same network as the host and receives network configuration from the physical router or DHCP server.

Role-Based Design Principles

The following principles govern all network configuration decisions:

- Network configuration depends on function, not operating system.
- Not all servers require multiple network adapters.
- Increase network complexity only when the role requires it.

Roles

- **Infrastructure/Network Services Machines:** These virtual machines provide core network services or act as traffic intermediaries for other machines. **Examples include**: Active Directory Domain Controllers, DHCP and DNS servers, routing or gateway servers, firewall or proxy servers, and VPN servers.
    - **Network Requirements:** Two network adapters are typically required:
        - **Internal Network Adapter:** Used for communication with client machines and provides DHCP, DNS, authentication, or directory services.
        - **External Network Adapter:** Used to reach external networks (Internet or host network) and can be configured as NAT or Bridged, depending on lab design.

      This dual-adapter design separates internal traffic from external connectivity, mirroring real-world enterprise network segmentation. It enables controlled communication between internal systems while allowing outbound access when required.
- **Application/Standalone Servers:** These virtual machines provide services but do not route traffic or provide core network infrastructure. **Examples include**: file servers, web servers, database servers, monitoring or logging servers, and application servers.
    - **Network Requirements:** One network adapter is sufficient, but attachment mode depends on access needs:
        - NAT for internet access and isolation
        - Internal Network, when communication is limited to lab systems

      These machines consume network services rather than provide them. Multiple adapters are unnecessary unless the server is explicitly designed to route or segment traffic.
- **Client Machines (User Endpoints)**: These virtual machines represent end-user systems. **Examples include:** Windows 11 desktop clients, Ubuntu Desktop installations, and test workstations.
    - **Default Network Requirements**: One network adapter attached to NAT.
    - **Use Cases**: Standalone testing, software installation, general learning and practice, and internet access through the host system.

    - **Lab-Specific Exception:** When a client machine must communicate with an internal lab server (e.g., Active Directory domain join):
        - Attach the client to the Internal Network
        - Do not attach NAT unless explicitly required
        - Internet access is optional and often unnecessary.

Configuration Steps

- Open VM Network Settings:
    - Power off the virtual machine
    - In Oracle VM VirtualBox Manager, right-click the VM, select Settings, and navigate to Network
- Configure Network Adapter(s). It is recommended to do this in 'Expert mode' for additional network adapters
    - Enable Adapter 1: Select the appropriate 'Attached to' mode (NAT, Internal Network, or Bridged).
- If applicable, configure the Internal Network: assign a clear network name (e.g., intnet-lab) and ensure all related virtual machines use the same name.
- Enable Adapter 2 if required: Attach to NAT (controlled external access) or Bridged Adapter (direct LAN access).
- Verify adapter Order: Infrastructure VMs typically use:
    - Adapter 1: Internal Network
    - Adapter 2: External Network

Operational Considerations

- Bridged networking exposes virtual machines directly to the physical network and should be used deliberately.
- NAT provides isolation and is recommended for most standalone and client systems.
- Internal networks are isolated by design and require routing to access external networks.
- Document network design decisions alongside the VM purpose to maintain clarity as environments grow.

Naming Conventions (Reference Guidance)

Consistent naming conventions improve clarity, manageability, and long-term usability of VM environments. While Oracle VM VirtualBox does not enforce naming standards, adopting simple and predictable naming practices helps administrators quickly identify VMs and understand their purpose.

Virtual Machine Name (VirtualBox Manager)

The virtual machine name is used only within the virtualization platform and is primarily for administrator reference. Recommended characteristics:

- Descriptive and human-readable
- May include spaces, OS name, and version
- Examples: Windows 11, Windows Server 2019, Ubuntu 24.04 Desktop.

Hostname (Guest OS)

The hostname identifies the system for networking and logging. They are subject to stricter formatting rules & should follow standard conventions:

- Lowercase, no spaces, hyphens preferred
- Short, descriptive, and role-aware
- Examples: win11-client, windows-server, ubuntu-client, ubuntu2404.

Post-Installation ISO and Boot Order

During VM installation, the operating system ISO is attached to the optical drive, and the boot order must be set to Optical first, then Hard Disk. The ISO must remain in its original location because VirtualBox references it by file path and does not copy it into the VM. After the operating system is installed and the VM boots successfully, the ISO is no longer required.

At this stage, empty the optical drive and update the boot order so that the hard disk is first, with the optical drive either listed after it or unchecked. This applies to both server and client VMs and prevents boot errors or "inaccessible" ISO warnings during normal operation.

Snapshot Management

Snapshot management creates a quick, saveable restore point so you can easily revert your virtual machine to its original state if something goes wrong in your lab. VirtualBox stores snapshot files directly on your host machine's drive, so make sure you have enough physical storage space first.

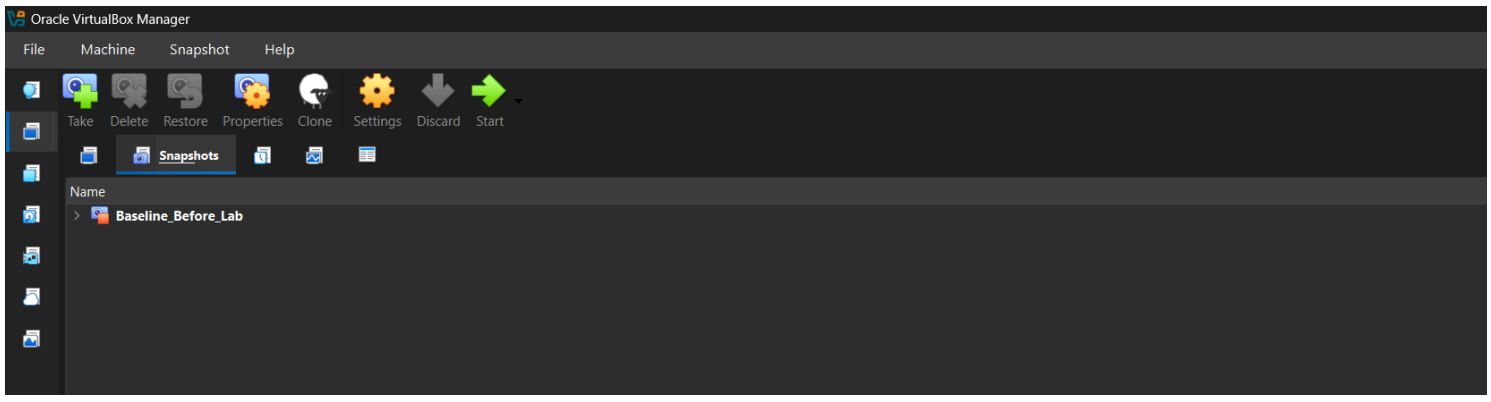
Step 1: Check Your Host Storage Space

VirtualBox snapshots save the "difference" between your original disk and changes you make during the lab. They grow larger the more you install or change things during the lab.

- Press Windows Key + E on your HP host to open File Explorer.
- Click This PC on the left menu.
- Ensure your Local Disk (C:) has at least 20 GB to 30 GB of free space before starting, as a heavy Windows lab can quickly fill up your drive.

Step 2: Take Your Pre-Lab Snapshot

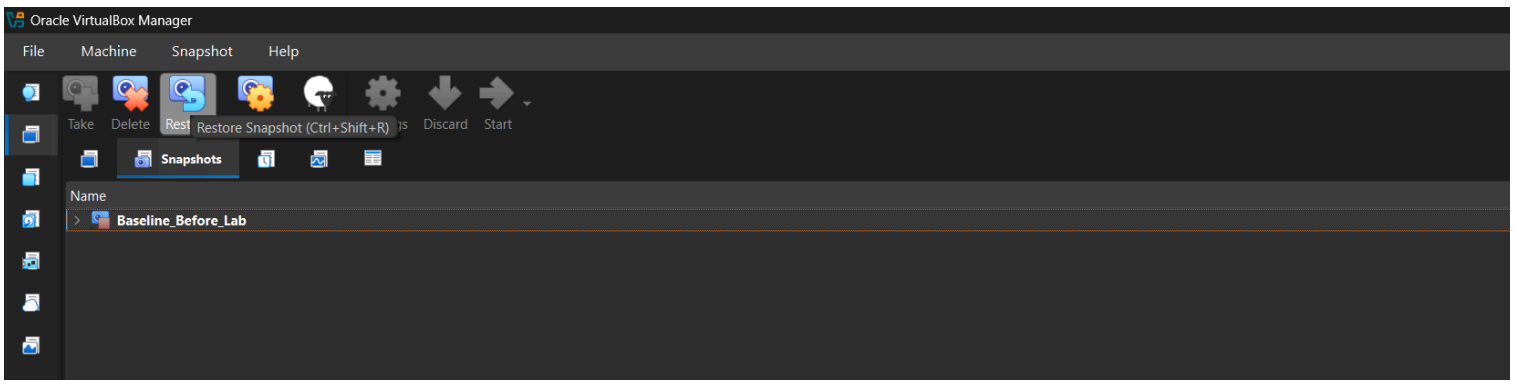
- Open VirtualBox Manager on your HP host.
- Fully Shut Down your VM (do not use "Save State", as powered-off snapshots are cleaner and more stable).
- Click on your VM in the left panel and switch to the Snapshots tab in the right panel.
- Click the Take button (camera icon) at the top of the snapshots window.
- Name it something clear, like *Baseline_Before_Lab,* and click OK.



Step 3: Reverting After Your Lab

If your lab breaks the VM, or if you simply want to wipe the slate clean to start over:

- Turn off the VM.
- Go back to the Snapshots section in VirtualBox.
- Click on your *Baseline_Before_Lab* snapshot and click the Restore button at the top.
- In the pop-up window, uncheck the box that asks to "Create a snapshot of the current machine state" (unless you want to save your broken lab work), then click Restore.



Step 4: Cleaning Up (Crucial for HP Drive Space)

Leaving snapshots active permanently slows down VM performance and eats up your HP host's hard drive space. Once your lab is finished and you no longer need to roll back:

- Shut down the VM.
- Select the snapshot you created (*Baseline_Before_Lab*).
- Click the Delete button (trash can icon) at the top.

VirtualBox will merge the data, saving your lab progress into the main VM file and freeing up the extra space on your host drive.

VM: Windows Server 2019 Standard (Desktop Experience)

Operating System Details

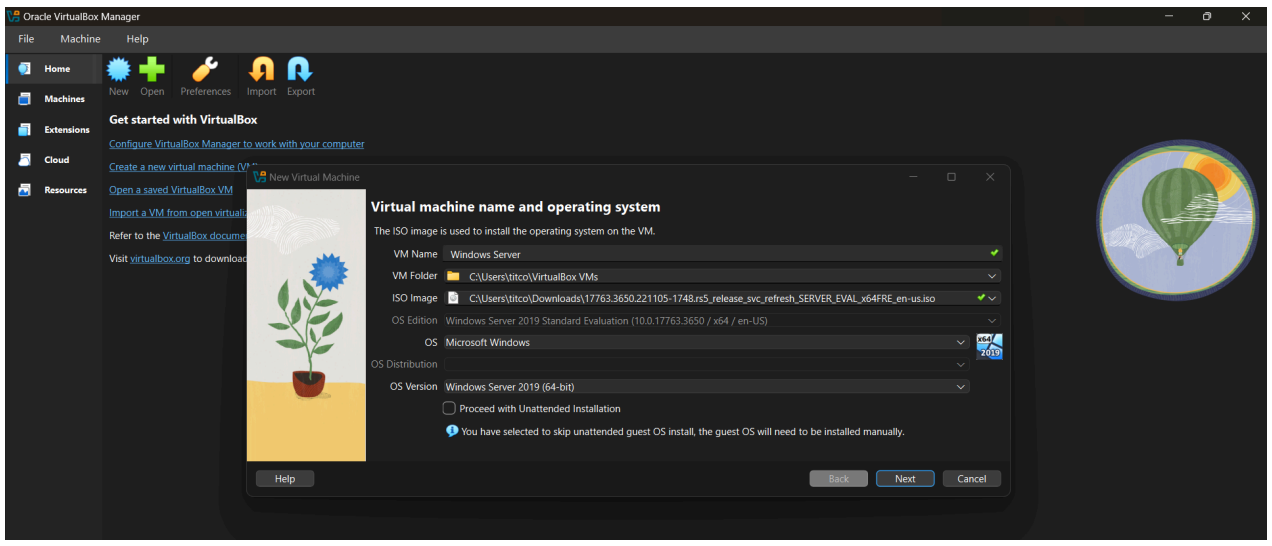
- **Name**: Microsoft Windows Server 2019 Standard Evaluation
- **Version**: 10.0.17763 Build 17763
- **System Type**: x64-based PC
- **ISO**: Windows Server 2019 Evaluation (180-day trial)

System Requirements

- Memory: 4 GB (4096 MB)
- CPU: 2 cores
- Storage: 80 GB
- Firmware: BIOS (EFI disabled)

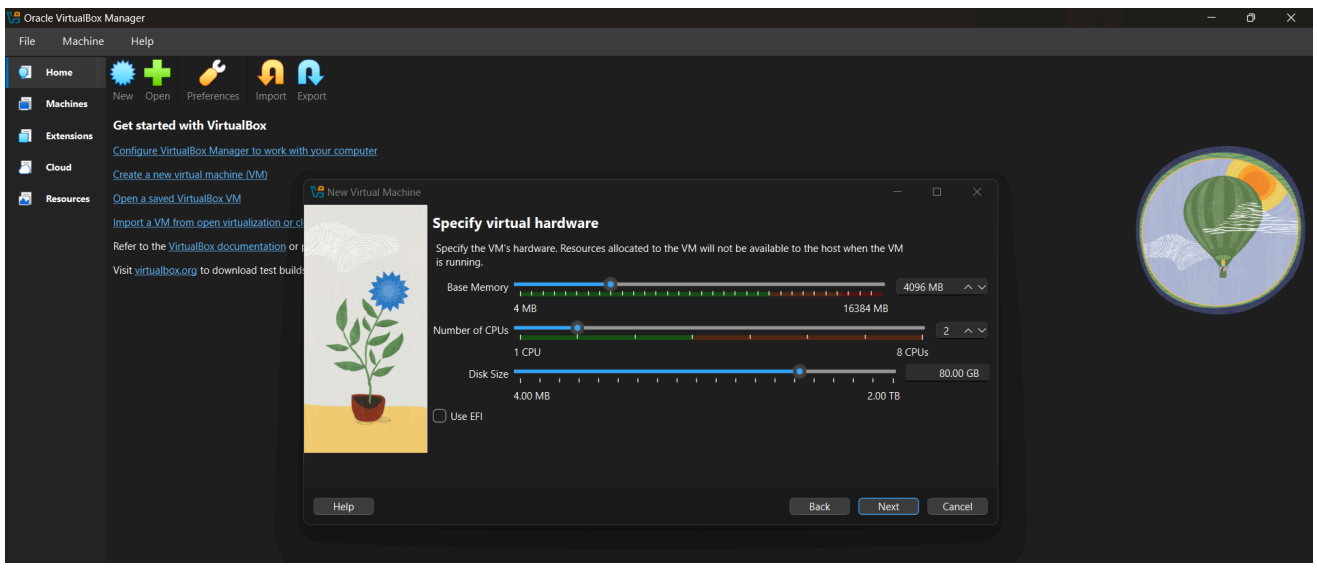
VM Creation

- Open Oracle VM VirtualBox (Home), click ‘New’ and enter:
    - Name: Windows Server
    - VM folder: Specify the directory where the virtual machine files will be stored
    - Attach the ISO image (In some cases, the OS edition and version will be automatically fetched; for others, they may have to be input).
- Disable ‘Unattended Installation’ for this VM.
    - Server installations often require closer control over setup steps, so a manual install is recommended.



Resource Allocation & Firmware Configuration

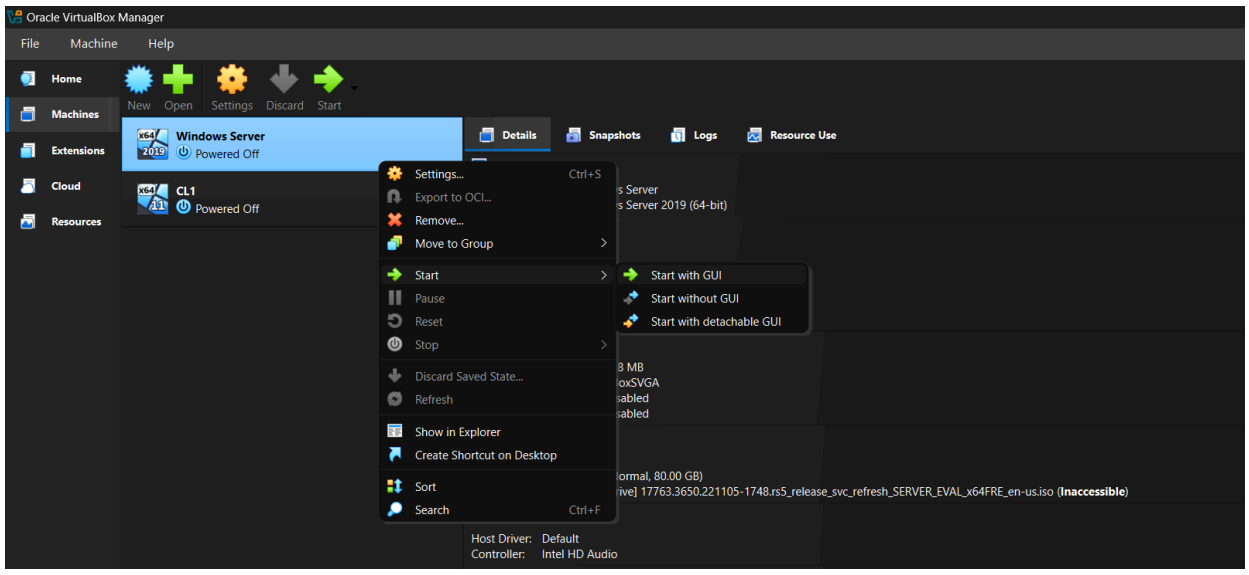
- Base Memory: 4096 MB
- Number of CPUs: 2 CPUs
- Disk Size: 80.00 GB
- EFI: Disabled
    - Use BIOS mode for stability and predictable boot behavior.



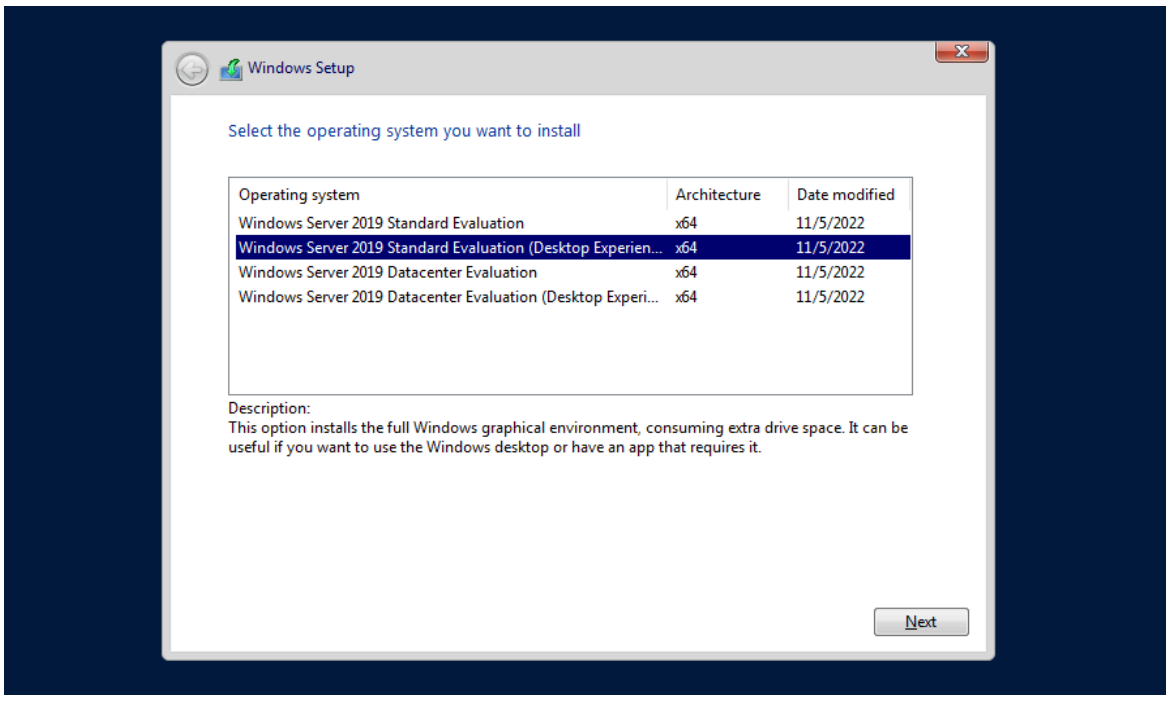
- Confirm VM creation details and click ‘Finish’.

Operating System Installation

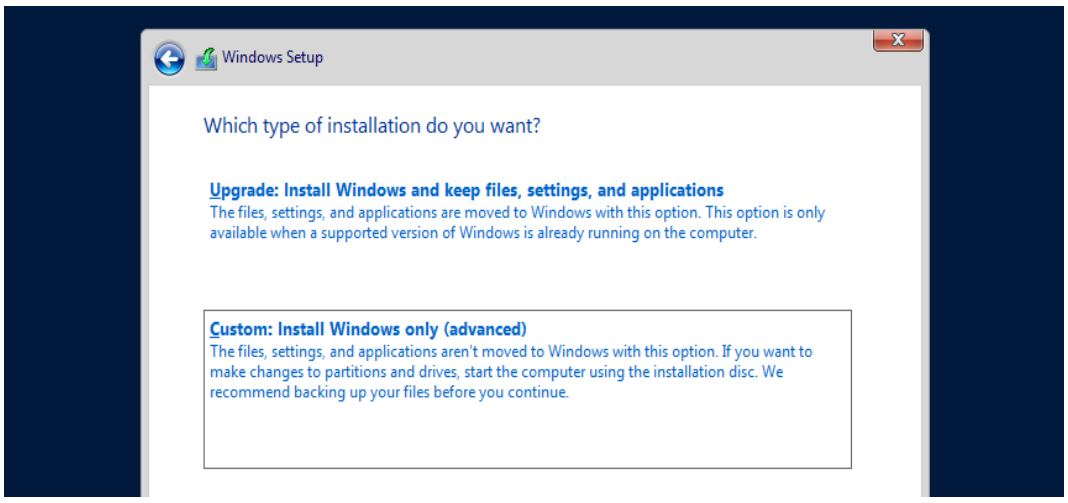
- Start the virtual machine: right-click on the VM, hover over the ‘Start’ option, and click on ‘Start with GUI.’



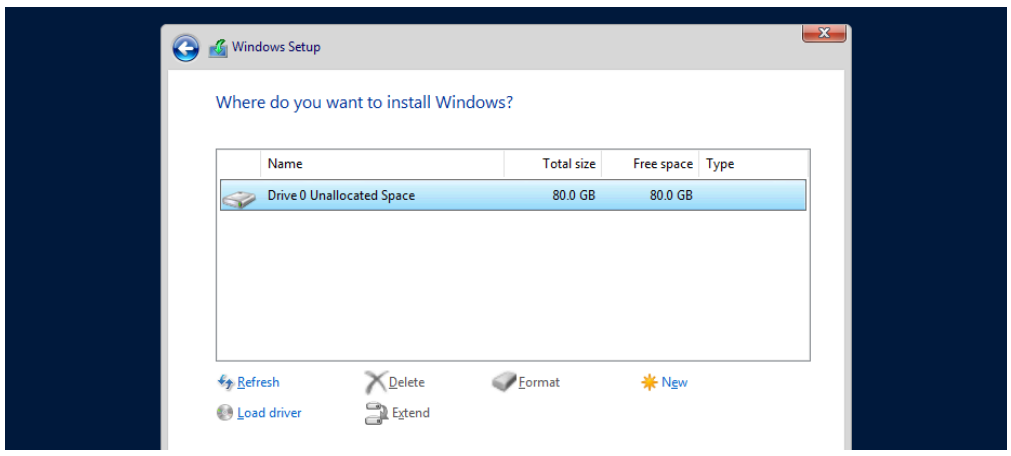
- Enter language & other preferences, then install Windows Server 2019.
- Select the operating system you want to install: choose Windows Server 2019 Standard (Desktop Experience).



- Accept the applicable notices and license terms.
- Choose installation type: select ‘Custom: Install Windows only (advanced)



- Select the install location: choose primary unallocated space on the disk.



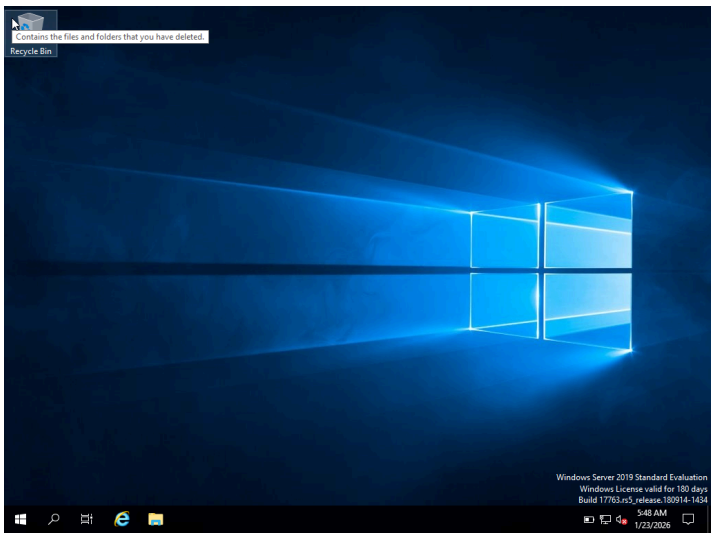
- Complete installation (this may take a while).
- Set administrator password and finish.

Initial Login

- Unlock machine: click ‘Input’, hover over the ‘keyboard’ option, and select ‘Insert Ctrl+Alt+Del’.

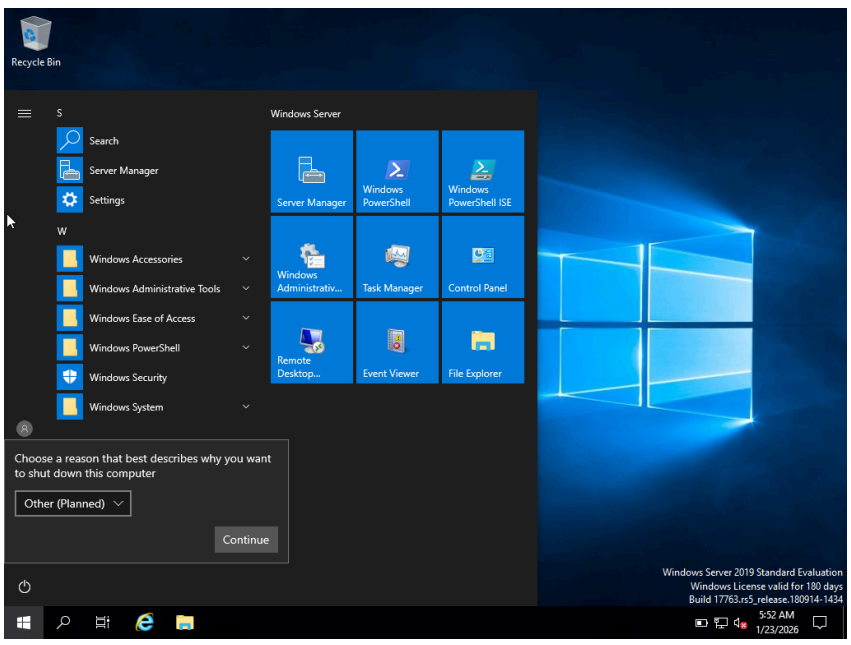


- Login: input ‘Administrator password ’. Successful login displays the Windows Server desktop.

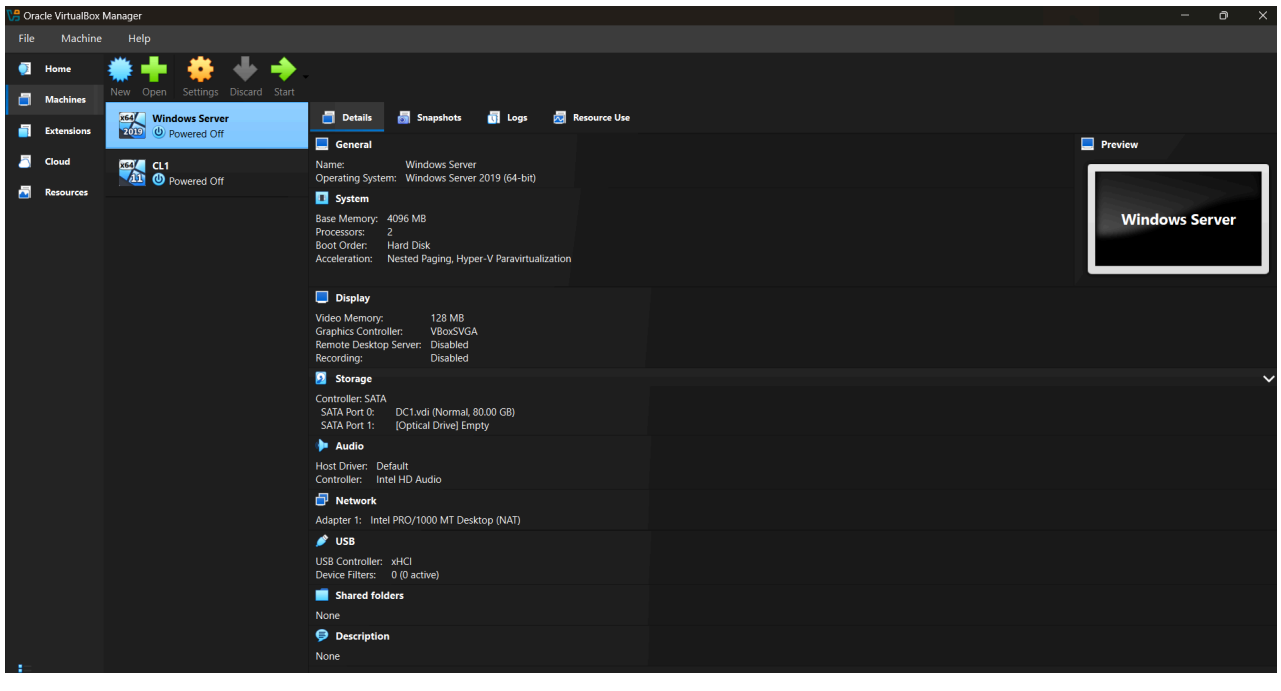


Shutting Down

- To power off the VM safely, click the Windows logo, then the power-off button, and select ‘Other (Planned)’ as the reason for shutting down.



Below are the VM settings. To configure those not included in the setup process, click the specific setting on the detail page. Not all settings are editable.



VM: Windows 11 Client (Unattended Installation)

Operating System Details

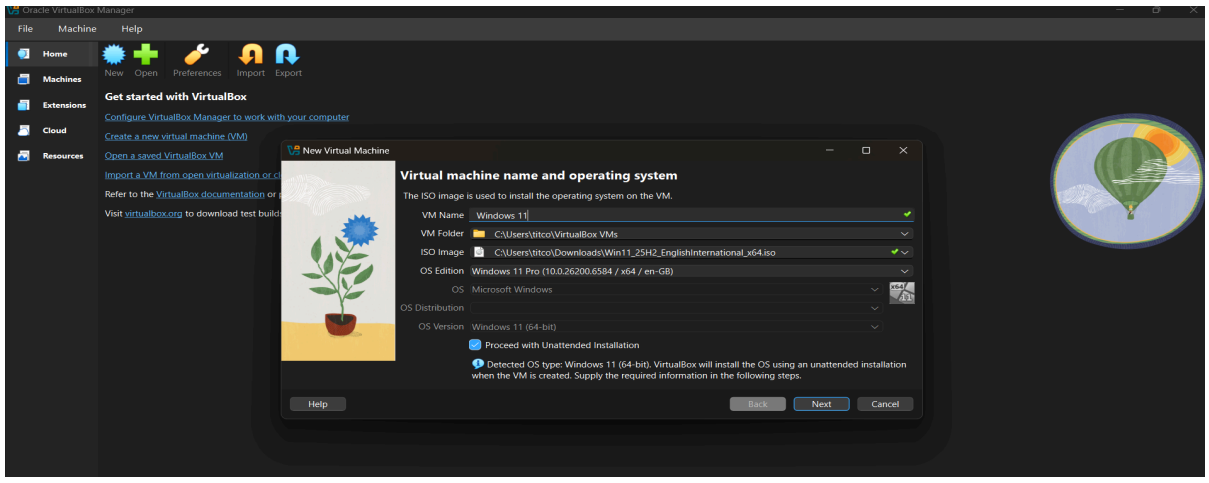
- **Name**: Microsoft Windows 11 Pro
- **Version**: 10.0.26200 Build 26200
- **System Type**: x64-based PC
- **ISO**: Windows 11 Pro

System Requirements

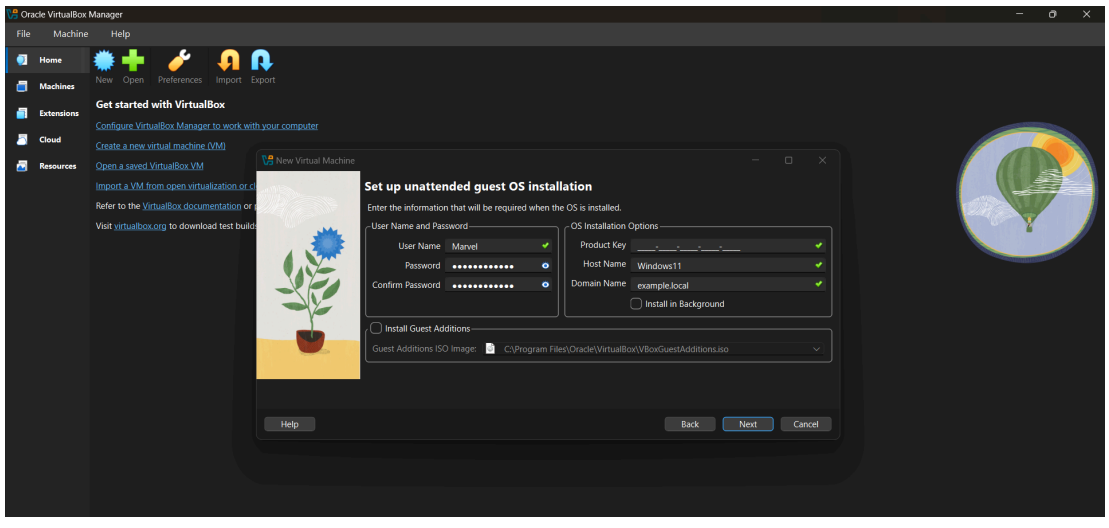
- Memory: 4 GB (4096 MB)
- CPU: 2 cores
- Storage: 80 GB
- Firmware: BIOS (EFI disabled)

VM Creation Process

- Open Oracle VM VirtualBox (Home), click 'New ', and enter:
    - Name: Windows 11
    - VM folder: Specify the directory where the virtual machine files will be stored
    - Attach the ISO image (In some cases, the OS edition and version will be automatically fetched; for others, they may have to be input).
- Enable 'Unattended Installation' for this VM.
    - Enabling unattended installation automates much of the setup and allows you to preconfigure user accounts and system identification.

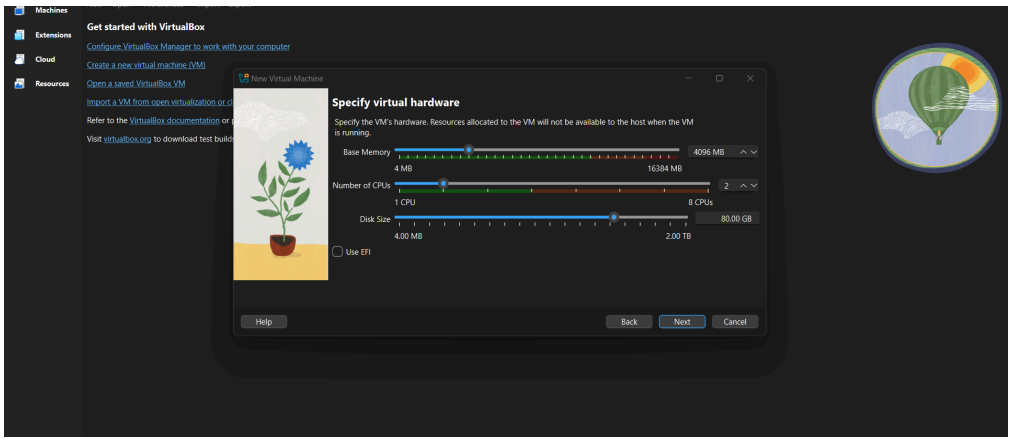


- Configure the following in the unattended guest OS setup:
    - Username: [your local account username]
    - Password: [your chosen password]
    - Host Name: [your VM hostname]
    - Domain Name: [your domain, e.g., techstars.local]



Resource Allocation & Firmware Configuration

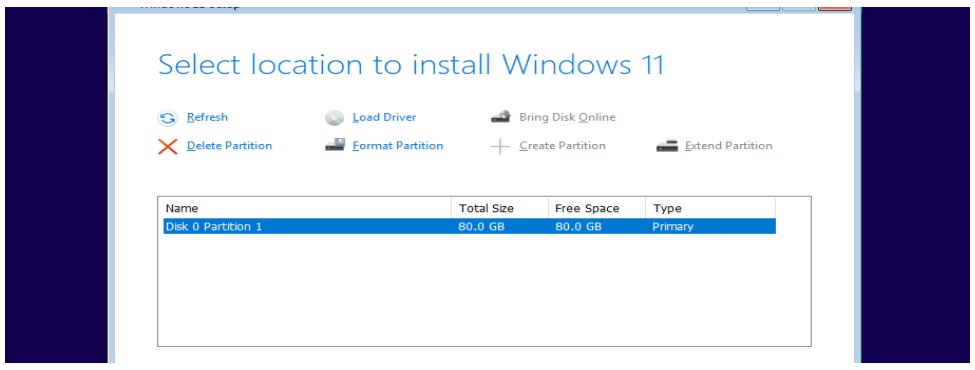
- Base Memory: 4096 MB
- Number of CPUs: 2 CPUs
- Disk Size: 80.00 GB
- EFI: Disabled
    - Use BIOS mode for stability and predictable boot behavior.



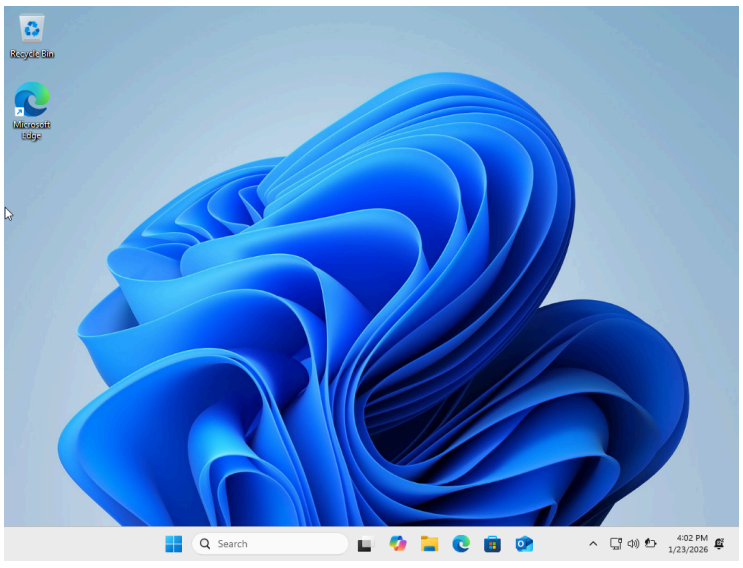
- Confirm VM creation details and click ‘Finish’. The VM will automatically boot.

Operating System Installation

- Select installation location:
    - The initial default disk may appear as Disk 0 Partition 1. Click ‘Delete Partition’ to remove the partition.



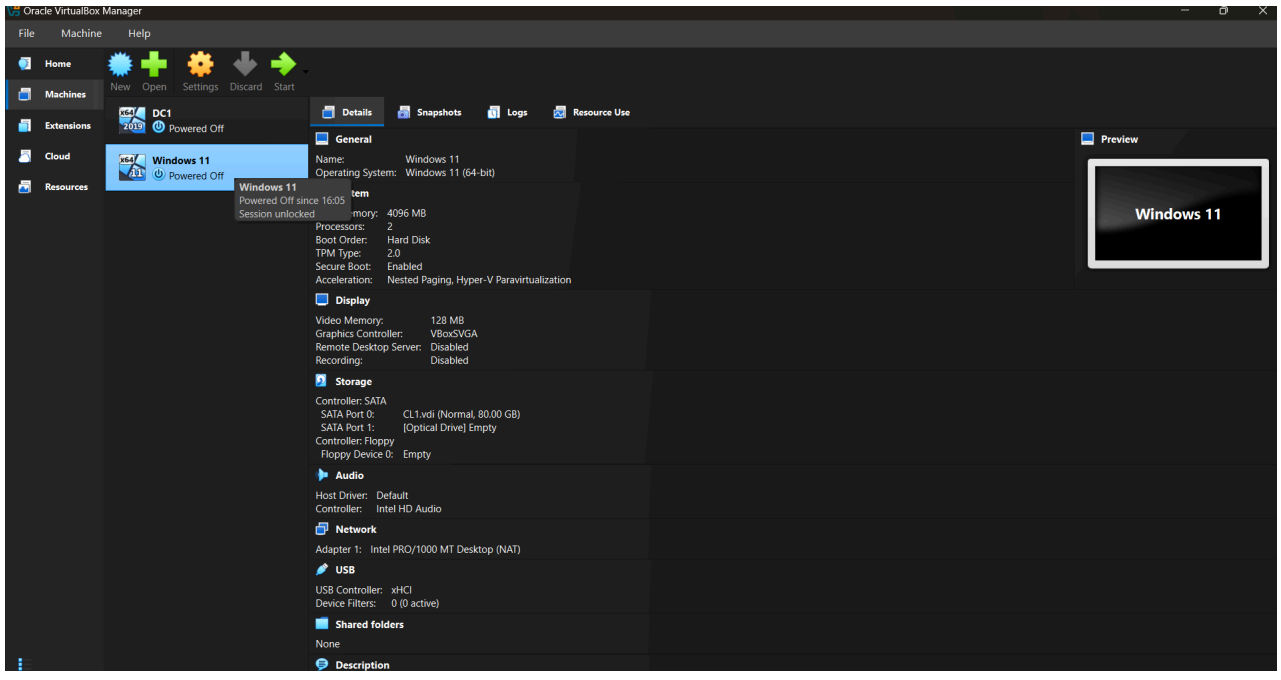
    - Afterwards, the disk will appear as Disk 0 Unallocated Space. Select this unallocated space for installation.
- Windows 11 installs automatically. The process may take several minutes. Then, the VM boots directly into the Windows 11 desktop.



Shutting Down

- To power off the VM safely, click on the Windows logo, then the power off button, and select ‘Shut down’.

Below are the VM settings. To configure settings not included in the setup process, click the specific setting on the detail page. Not all settings are editable.



VM: Kali Linux 2026.2 Desktop (attended Installation)

Operating System Details

- **Name**: Kali 2026.2
- **OS Distribution**: Kali Linux
- **System Type**: amd64-based PC
- ISO: Kali Linux 2026.2

System Requirements

- Memory: 6 GB (6144 MB)
- CPU: 2 cores
- Storage: 25 GB

ISO Integrity Verification

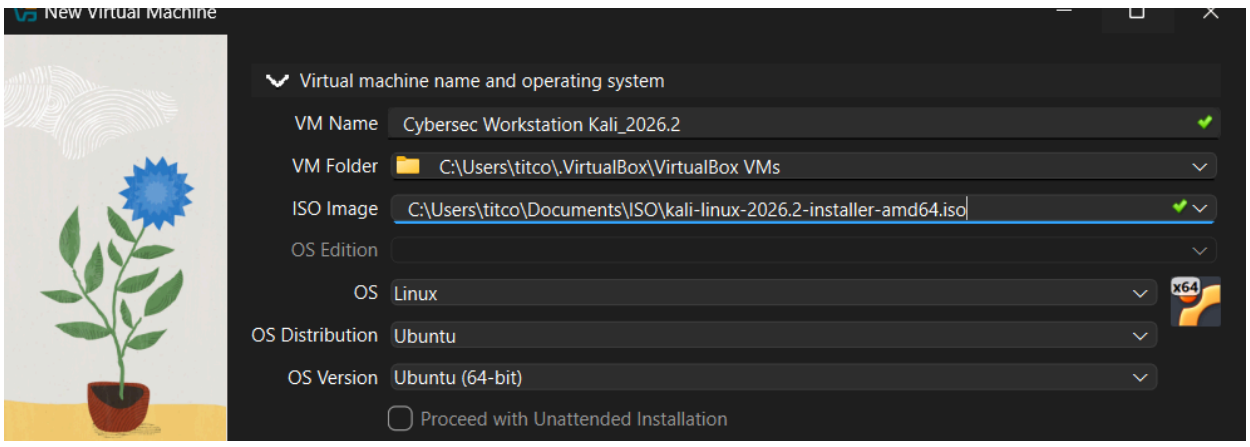
After downloading, verify that the SHA256 checksum of the downloaded Kali Linux ISO matches the official hash published on the Kali Linux website.

```
6dbefacc95e3b556c19c48e8bae39b8b505e2d3a1aba0bfb7ab62b036c3d2ba3  kali-linux-2026.2-installer-amd64.iso
```

```
C:\Users\titco>certutil -hashfile C:\Users\titco\Documents\ISO\kali-linux-2026.2-installer-amd64.iso SHA256
SHA256 hash of C:\Users\titco\Documents\ISO\kali-linux-2026.2-installer-amd64.iso:
6dbefacc95e3b556c19c48e8bae39b8b505e2d3a1aba0bfb7ab62b036c3d2ba3
CertUtil: -hashfile command completed successfully.
```

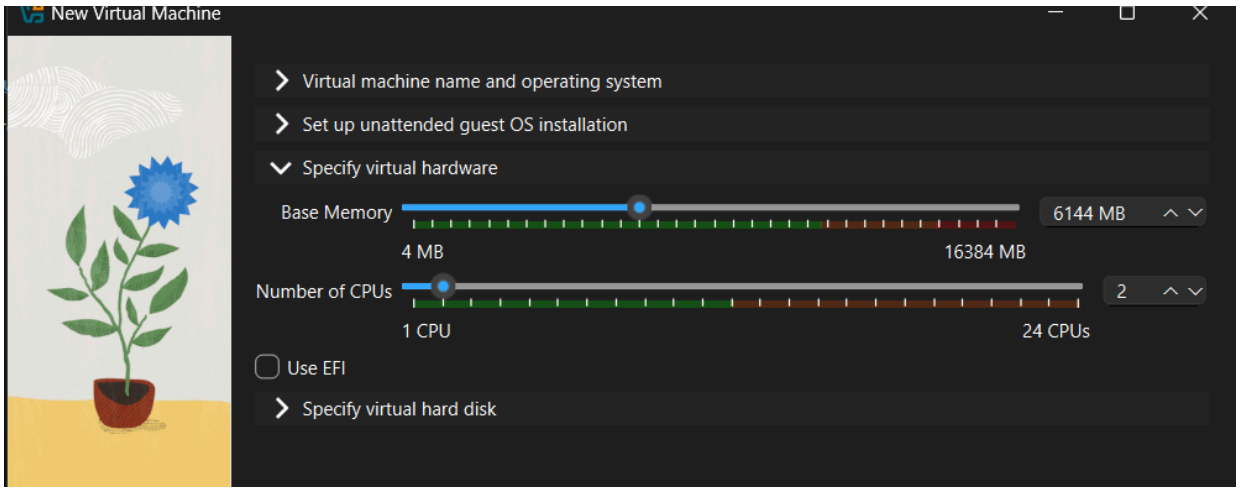
VM Creation Process

- Open Oracle VM VirtualBox (Home), click ‘New ’, and enter:
  - Name: Cybersec Workstation 24.04.5 LTS
  - VM folder: Specify the directory where the virtual machine files will be stored
  - Attach the ISO image
- Leave ‘Proceed with Unattended Installation’ unchecked.
  - Attended installation allows you to manually set up and configure the OS (including user accounts and system identification).



Resource Allocation & firmware configuration

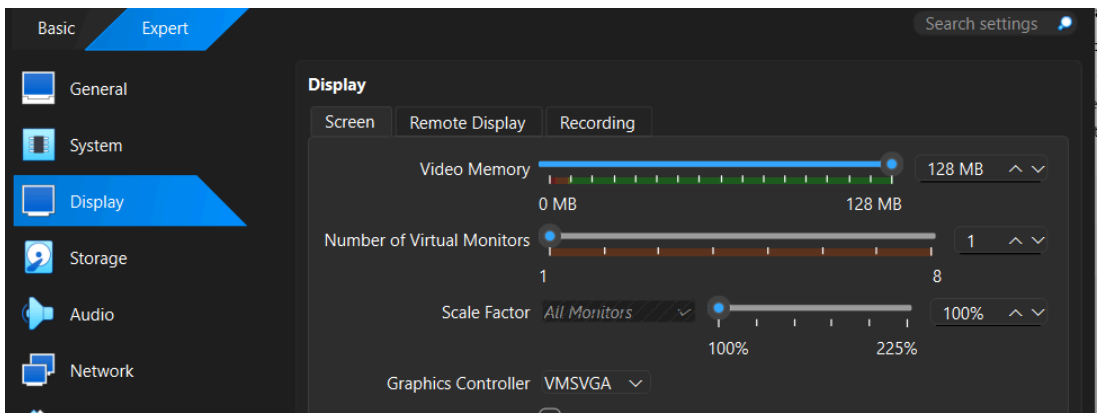
- Base Memory: 6144 MB
- Number of CPUs: 2 CPUs
- EFI: Disabled
  - Use BIOS mode for stability and predictable boot behavior.



- Disk Size: 50.00 GB
    - Do not pre-allocate the full size. Dynamic allocation lets the VM grow without using the host's storage upfront.
- Confirm VM details and click 'Finish'.

Additional Settings

- Click the newly created VM and select the settings icon at the top of the screen (Machines). Under 'Expert':
    - Network Adapter: Set to NAT (Network Address Translation).
    - General: Set both 'Shared Clipboard' and 'Drag-and-Drop' features to Bidirectional. This lets you use the shared clipboard from the host machine into the VM and drag-and-drop between them.
    - Display: Increase video memory to 128 MB for better display.



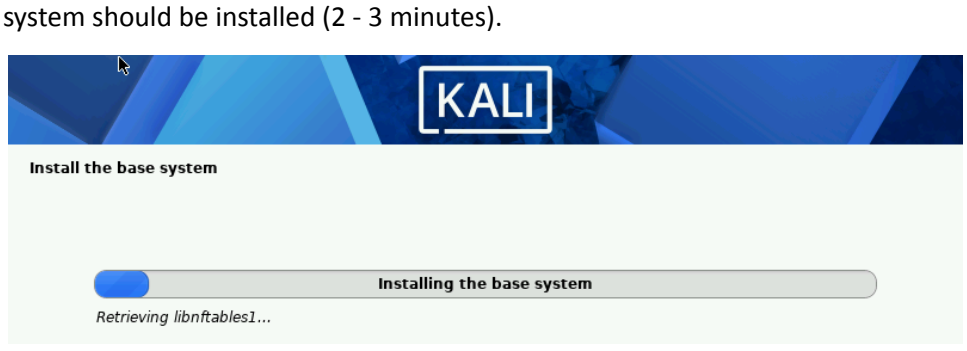
    - Save to apply the settings. Then, start the VM.

Operating System Installation

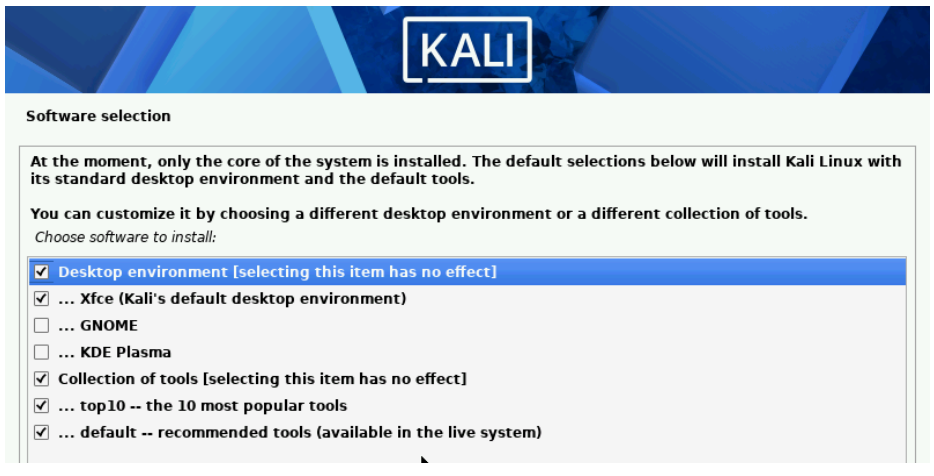
- On first boot, select the 'Graphical Install' option.



- Go through the setup prompts and configure the OS, including:
    - Name: analyst
    - Host Name: [your VM hostname]
    - Password: [your chosen password]
    - Domain Name: [your domain, e.g., cybersec.lab]
- After setup, the base system should be installed (2 - 3 minutes).



- After installation, select the desktop environment software and default tools, and continue.



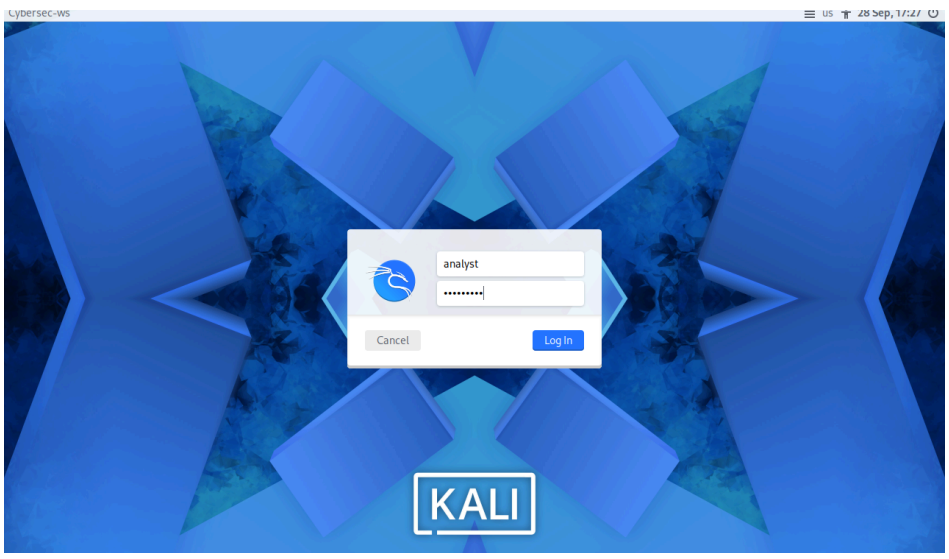
- The software should begin installating (20 - 30 minutes).



- Install the GRUB boot loader. Select the full name of the primary drive.
- The OS should finish installing (10-15 minutes).
- Press ‘Continue’ to reboot. This empties the Optical Drive, allowing the OS to boot from the Hard Disk. You can check that the Boot Order adjusts accordingly in the VM description in VirtualBox.

Initial Login

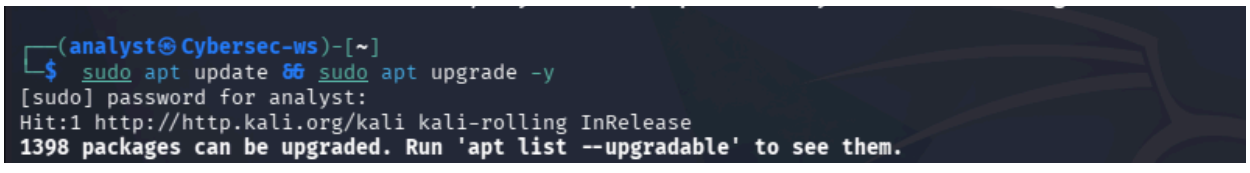
- The VM should restart. Input your login details and access the desktop.



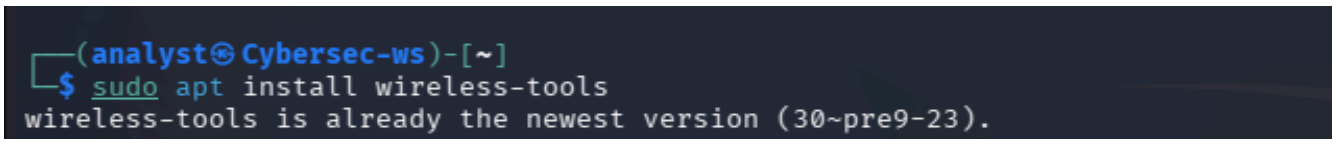


Post-Installation Configuration

- After initial login, update all pre-installed packages to their latest stable versions using the terminal.



- Install wireless tools for network interface configuration from the terminal.



Shutting Down

- To power off the VM safely, click on the X icon at the top-right corner of the screen and select ‘Power off the machine’.